

Task Division — Talk-to-my-Data

Artefact Capstone 2026 / Case #1: Building a real-world Generative AI application

Rotation across all 4 pillars: everyone takes a different sub-task each pillar, so by Stage 3 everyone has touched data, AI, backend, and frontend. Tasks marked Required come directly from Artefact's brief — don't skip these. Tasks marked Stretch are what makes the project stand out, but are the first things to cut if the timeline gets tight.

Note: Stage numbering and pillar numbering are separate — Stage 0's tasks are 0.1-0.4, Stage 1's are 1.1-1.3, and each pillar's tasks restart at .1 within that pillar.

Stage 0 — Prerequisites ☒ COMPLETE

Sub-task	Owner	Priority	Deadline
0.1 Cloud Setup (GCP project, CLI, service decisions) ☒ DONE	Raed	Complete	Closed
0.2 Environment Setup (Python, venv, IDE, .gitignore) ☒ DONE	Muhammed	Complete	Closed
0.3 Architecture Sketch v1 (Excalidraw) ☒ DONE	Amir	Completed by Raed	Closed
0.4 Dataset & KPI Draft ☒ DONE	Fares	Complete	Closed

Gate cleared: GCP project verified working end-to-end, environment documented, architecture sketch presented to Mark + Quinten, dataset downloaded and KPIs defined.

Stage 1 — Contracts & Skeleton ☒ COMPLETE

Sub-task	Owner	Priority	Deadline
1.1 Write schema.md ☒ DONE	All (together)	Complete	Closed
1.2 Write api.md ☒ DONE	All (together)	Complete	Closed
1.3 Repo Skeleton (/src, /mock_data, /docs) ☒ DONE	All (together)	Complete	Closed

Gate cleared: both contract files committed and rendering correctly, folder skeleton matches the plan.

Stage 2 — Core Build

Pillar 1 — Cloud & Data Platform COMPLETE

Sub-task	Owner	Priority	Deadline
1.1 Retail Data Cleaning & ETL  DONE	Raed	Complete	Closed
1.2 BigQuery Data Warehouse Setup  DONE	Fares	Complete	Closed

Gate cleared: real data queryable in BigQuery by everyone.

Note: 1.3 (IAM & Security) and 1.4 (Docker & Cloud Run) moved out of this pillar -- see Stage 3 and Stage 5.

Pillar 2 — LLM & Text-to-SQL

Sub-task	Owner	Priority	Deadline
2.1 Dynamic Schema Injection  DONE	Raed	Complete	Closed
2.2 Few-Shot Prompting & Domain Lexicon  DONE	Amir	Completed by Raed	Sat, Sep 26
2.3 Self-Healing Query Feedback Loop  DONE	Muhammed	Complete	Closed
2.4 Ambiguity Resolution & Clarification  DONE	Fares	Complete	Closed

Gate: a plain-English question reliably produces correct SQL against real data — met by 2.1/2.3/2.4. 2.2 remains open as a stretch enhancement and does not block Pillar 3/4.

Pillar 3 — Backend & Analytics Engine

Sub-task	Owner	Priority	Deadline
3.1 FastAPI Core Service Layer  DONE	Raed	Complete	Sun, Sep 27
3.2 SQL Validation & Safety Guardrails 	Muhammed	Complete	Sun, Sep 27
3.3 Retail KPI Computation Engine 	Amir	Completed by Raed	Sun, Sep 27
3.4 Multi-Modal Response Formatter  DONE	Fares	Complete	Sun, Sep 27

Gate: /ask endpoint works end-to-end, callable directly, no UI needed yet.

Pillar 4 — Frontend UI, Benchmark & Delivery

Sub-task	Owner	Priority	Deadline
4.1 Streamlit Conversational Interface	Muhammed	Required	Sun, Oct 4
4.2 Dynamic Data Visualizations	Raed	Stretch	Sun, Oct 4
4.3 Golden Benchmark Evaluation Suite	Fares	Stretch (10-15 questions is fine)	Sun, Oct 4
4.4 Excalidraw Architecture & Client Deck	Amir	Required — named in the brief	Sun, Oct 4

Gate: a real question typed into the real UI returns a real, working answer.

Stage 3 — Integration

Sub-task	Owner	Priority	Deadline
Full Pipeline Integration (replace remaining mocks, test end-to-end)	All (together)	Required	Sun, Oct 4
1.4 Docker & Cloud Run Deployment (moved here from Pillar 1)	Muhammed	Required	Sun, Oct 4

Gate: the full system works start to finish, without anyone manually patching things together.

Stage 4 — Dry Run Prep

Sub-task	Owner	Priority	Deadline
Dry Run Deck + Rehearsal, presented to Mark + Quinten	All (together)	Required	Mon, Oct 5

Gate: feedback received, clear list of what to fix before the final.

Stage 5 — Final Polish & Deployment

Sub-task	Owner	Priority	Deadline
Full Cloud Run Deployment + Final Deck + Live Demo Rehearsal	All (together)	Required	Sun, Oct 18
1.3 IAM & Security Configuration (deferred from Pillar 1)	Amir	Stretch	Sun, Oct 18

Gate: live demo works, deck is ready, everyone knows their speaking part.